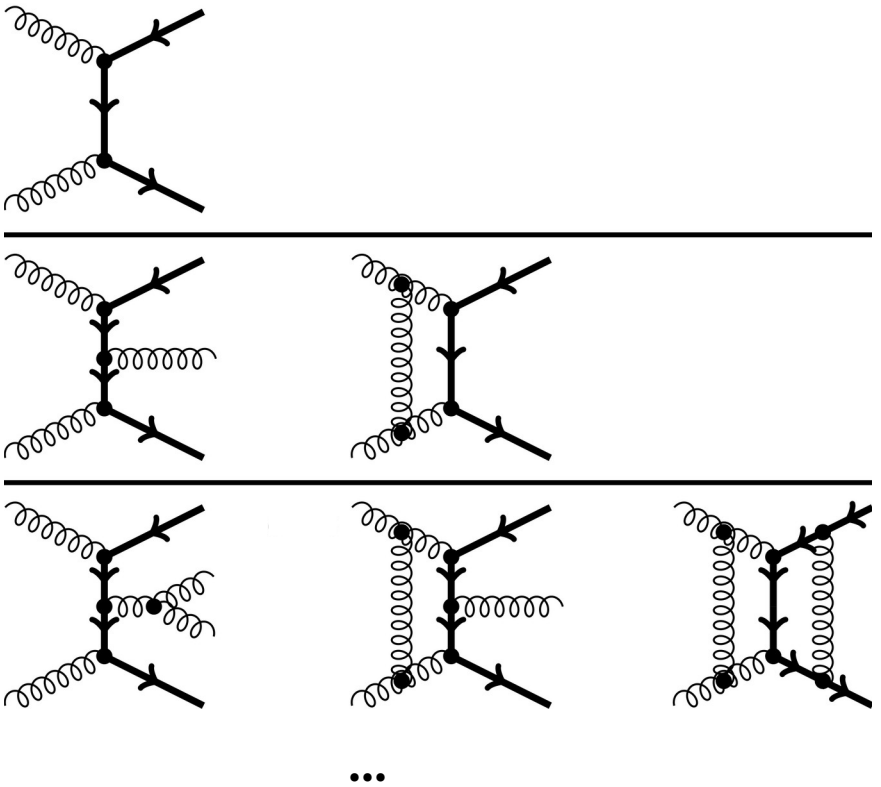
# Quantum Chromodynamics

René Poncelet

Lecture 2:
initial state partons & PDFs, e+e- @ NLO QCD

THE HENRYK NIEWODNICZAŃSKI
INSTITUTE OF NUCLEAR PHYSICS
POLISH ACADEMY OF SCIENCES

**European School of High Energy Physics**
19th June - 3rd Jul 2026 - Ostroda



# Lecture outline

- Motivation
- The QCD Lagrangian
  + running of the strong coupling
- Infrared divergencies
- **Parton-distribution functions**
- **Anatomy of a higher-order QCD computation**
  - **Virtual/Loop corrections**
  - **Real emission corrections**
- Systematics of higher-order computations
- Phenomenology

# Lecture 1 - recap

- A bit of history…
- Structure of QCD as a SU(3) local gauge theory
- UV Renormalization
- Running of the strong coupling constant
- Soft/collinear divergencies for gluon emissions

# Hadrons & partons in the initial state

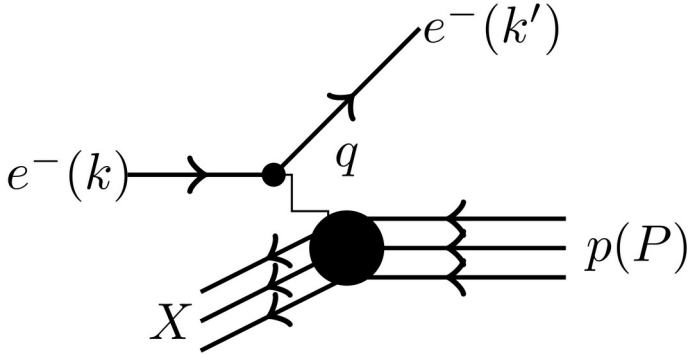
# Deep inelastic scattering

Consider striking a hadron with a high energy electron
(forget about parton model of hadrons for a moment)

→ assume we only measure the outgoing electron momentum

+ control over initial state electron + proton momenta

$$e(k) + p(P) \to e(k') + X$$




Some useful kinematic quantities:

$$s = (P+k)^2 \quad q^\mu = k^\mu - k^{\mu\prime} \quad Q^2 = -q^2$$

$$x = \frac{Q^2}{2P \cdot q} \qquad y = \frac{P \cdot q}{P \cdot k}$$

# Deep-inelastic scattering cross sections

Cross-section: $$\mathrm{d}\sigma = \sum_X \frac{1}{4m_p E} \int \mathrm{d}\Phi \frac{1}{4} \sum_{\mathrm{pol}} |\mathcal{M}|^2$$

$$\mathrm{d}\Phi = \frac{\mathrm{d}^3\vec{k}'}{2E'(2\pi)^2} \mathrm{d}\Phi_X$$

Decomposition into Leptonic & Hadronic current:

$$\frac{1}{4} \sum_{\mathrm{pol}} |\mathcal{M}|^2 = \frac{e^4}{Q^4} L^{\mu\nu} H_{\mu\nu}$$

Exploit this factorization and integrate out X: $$W_{\mu\nu} = \frac{1}{8\pi} \sum_X \int \mathrm{d}\Phi_X H_{\mu\nu}$$

→ W should only depend on P and q now

Decompose tensor structure into P and q:

$$W_{\mu\nu}(P,q) = \left(-g_{\mu\nu} + \frac{q_\mu q_\nu}{q^2}\right) W_1(x,Q^2) + \left(P_\mu - q_\mu \frac{P\cdot q}{q^2}\right)\left(P_\nu - q_\nu \frac{P\cdot q}{q^2}\right) \frac{W_2(x,Q^2)}{P\cdot q}$$

# DIS structure functions

Contract with lepton current:

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4}\left[y^2 W_1(x,Q^2) + \left(\frac{1-y}{x} - xy\frac{m_p^2}{Q^2}\right) W_2(x,Q^2)\right]$$

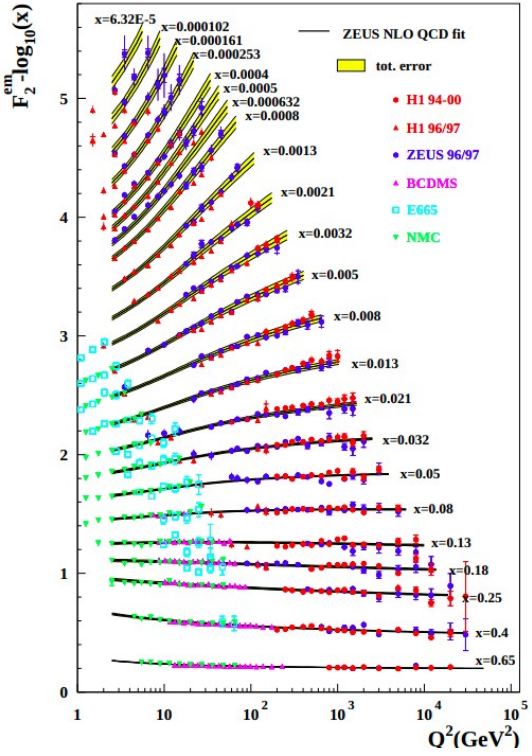
Scaling limit: $\quad Q^2 \to \infty\,, \quad x = \text{const.}$

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4}\left[(1+(1-y)^2)F_1(x,Q^2) + \frac{1-y}{x}(F_2 - 2xF_1)\right]$$

$$F_2(x,Q^2) \equiv F_2(x)$$

$$F_2(x) = 2xF_1(x)$$

Callan-Gross relation (scattering from elementary ½ particles)

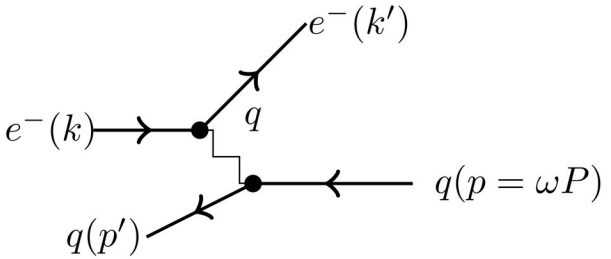



In reality we observe violations of this scaling law at small-x → saturation

# Parton model of the proton

Let's assume scattering off an quark with a fraction of the proton momentum

$$p^\mu = \omega P^\mu \qquad e(k) + q_f(p) \to e(k') + q_f(p')$$




$$p^2 = p'^2 = (p+q)^2 = p^2 + Q^2 + 2\omega Pq \;\to \omega = \frac{Q^2}{2P\cdot q} = x$$

A few changes on the previous formula:

$$\mathrm{d}\sigma = \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_2 \frac{1}{4} \sum_{\mathrm{pol}} |\mathcal{M}|^2$$

$$\frac{1}{4}\sum_{\mathrm{pol}} |\mathcal{M}|^2 = \frac{e^4 e_f^2}{Q^4} L^{\mu\nu} Q_{\mu\nu}$$

Here we can compute the hadronic part, in first order perturbation theory: (compare to yesterdays e+e- example)

$$Q^{\mu\nu} = p^\mu p^{\nu\prime} + p^\nu p^{\mu\prime} - g^{\mu\nu}(p\cdot p')$$

$$\frac{1}{4}\sum_{\mathrm{pol}} |\mathcal{M}|^2 = \frac{2e_f^2 e^4 \hat{s}}{Q^4}(1 + (1-y)^2)$$

# Quark-electron cross section

Phase space parameterization:

$$\mathrm{d}\Phi_2 = \frac{\mathrm{d}^3k'}{(2\pi)^3 2E'}\frac{\mathrm{d}^4p'}{(2\pi)^3}\delta(p'^2)(2\pi)^4\delta^4(k+p-k'-p') = \frac{\mathrm{d}\phi}{(4\pi)^2}\mathrm{d}x\mathrm{d}y\delta(\omega-x)$$

Integrating out the azimuth and simplify:

Quark: $$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4}\left[1+(1-y)^2\right]\frac{1}{2}e_f^2\delta(\omega-x) \qquad +0$$

Hadron: $$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4}\left[(1+(1-y)^2)F_1(x,Q^2)+\frac{1-y}{x}(F_2-2xF_1)\right]$$

… found Callan-Gross relation!

if we interprete a hadron as bunch of quarks…

# Parton model of the proton

Introduce the parton distribution function: $f_i(x)$
Propability to find a parton of flavour $i$ with momentum fraction $x$

$$F_2(x) = \sum_i \int \mathrm{d}\omega f_i(\omega) x e_{f_i}^2 \delta(x - \omega) = \sum_i e_{f_i}^2 f_i(x) x$$

Re-express differential cross section in terms of PDFs:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}x\mathrm{d}Q^2} = \int_x^1 \frac{\mathrm{d}\omega}{\omega} \sum_i f_i(\omega) \frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}x\mathrm{d}Q^2}(\frac{x}{\omega}, Q^2)$$

$t_{\mathrm{had}} \sim 1/m_p$

$Q \gg m_p \leftrightarrow t_{\mathrm{hard}} \ll t_{\mathrm{had}}$

'hadronic' cross section

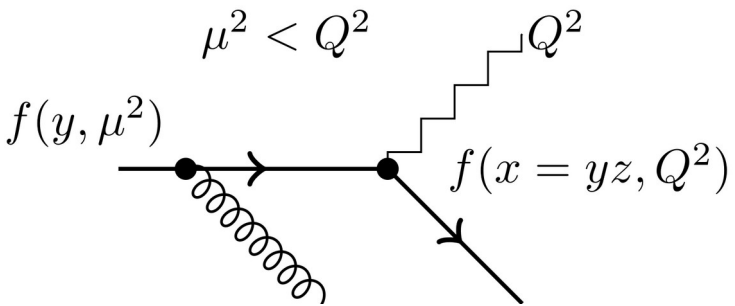
'partonic' cross section

The quarks inside a proton are not inert, they do interact
→ what if they emit gluons?
→ what is the impact on the parton-distribution functions?

# Evolution of PDFs

The PDFs depend on the scale they are probed at $f_{a,h}(x, \mu_F^2)$

→ the origin are emissions at a lower energy scale
consider striking a parton with a virtual photon $Q^2$




splitting functions!

$$f(x, Q^2) = f(x, \mu^2) + \int_x^1 \mathrm{d}y f(y, \mu^2) \int_0^1 \mathrm{d}x \delta(x - yz) \int_{\mu^2}^{Q^2} \mathrm{d}k^2 P(z, k^2)$$

no emission

all possibilities of emissions

# DGLAP

$$f(x,Q^2) = f(x,\mu^2) + \int_x^1 \mathrm{d}y f(y,\mu^2) \int_0^1 \mathrm{d}x \delta(x-yz) \int_{\mu^2}^{Q^2} \mathrm{d}k^2 P(z,k^2)$$

$$\longrightarrow \quad \frac{\mathrm{d}f(x,\mu^2)}{\mathrm{d}\mu^2} = \int_x^1 \mathrm{d}z P(z,\mu^2) f(x/z,\mu^2)$$

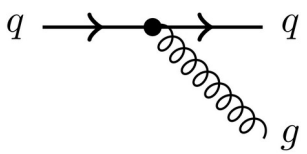
Taking into account that the splitting may change the parton flavour:
DGLAP [Dokshitzer, Gribov, Lipatov, Altarelli, Parisi, 77]

$$\frac{\mathrm{d}f_a(x,\mu^2)}{\mathrm{d}\mu^2} = \int_x^1 \mathrm{d}z \frac{\alpha_s}{2\pi} P_{ab}(z,\mu^2) f_b(x/z,\mu^2)$$

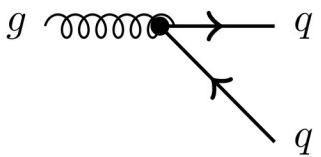
Prediction of the running of PDFs (universal, i.e. not dependent on the collider)
→ still needs input at a given scale (similar to coupling)

# DGLAP: splitting functions

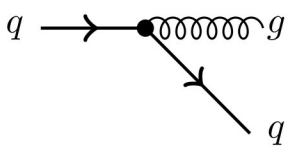
$$\frac{\mathrm{d}f_a(x,\mu^2)}{\mathrm{d}\mu^2} = \int_x^1 \mathrm{d}z \frac{\alpha_s}{2\pi} P_{ab}(z,\mu^2) f_b(x/z,\mu^2)$$



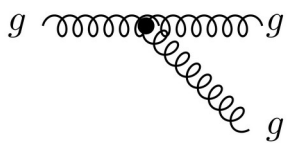

$$P_{qq}(z) = C_F \left[ \left( \frac{1+z^2}{1-z} \right)_+ + \frac{3}{2}\delta(1-z) \right]$$




$$P_{qg}(z) = T_F(z^2 + (1-z)^2)$$




$$P_{gq}(z) = C_F \left( \frac{1+(1-z)^2}{z} \right)$$



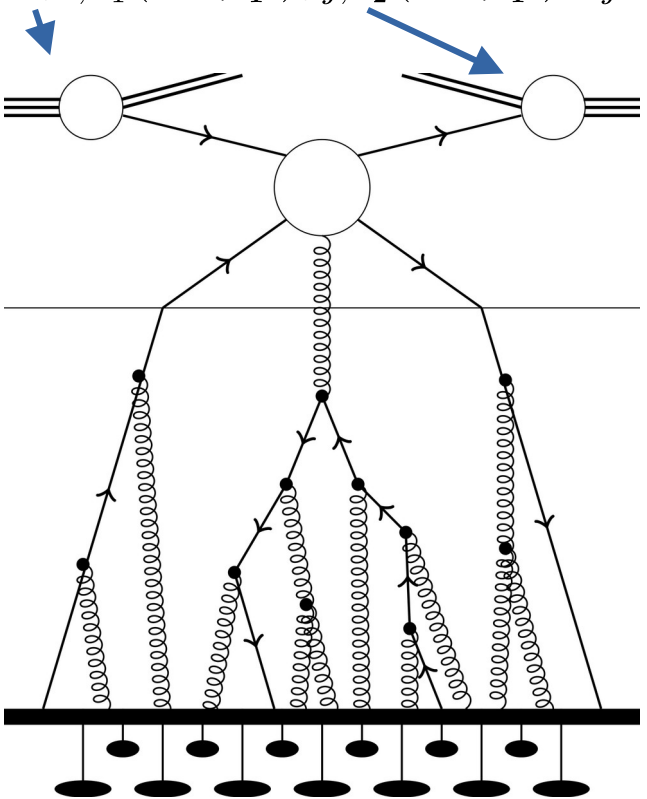

$$2C_A \left( \left( \left( \frac{z}{1-z} \right)_+ + \frac{1-z}{z} + z(1-z) \right) + \delta(1-z) \left( \frac{11N_c - 2n_f}{6} \right) \right.$$

'+' distribution:

$$\int_0^1 \mathrm{d}x (f(x))_+ g(x) = \int_0^1 \mathrm{d}x f(x)(g(x) - g(1))$$

# Hadron-hadron collisions

$$\sigma_{h_1 h_2 \to X} = \sum_{ij} \int_0^1 \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 f_{i,h_1}(x_1, \mu_F^2) f_{j,h_2}(x_2, \mu_F^2) \hat{\sigma}_{ij \to X}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2) + \mathcal{O}\left(\frac{\Lambda^2}{Q^2}\right)$$



Crucially: **PDFs are universal**
i.e. independent of the process
→ same as in DIS

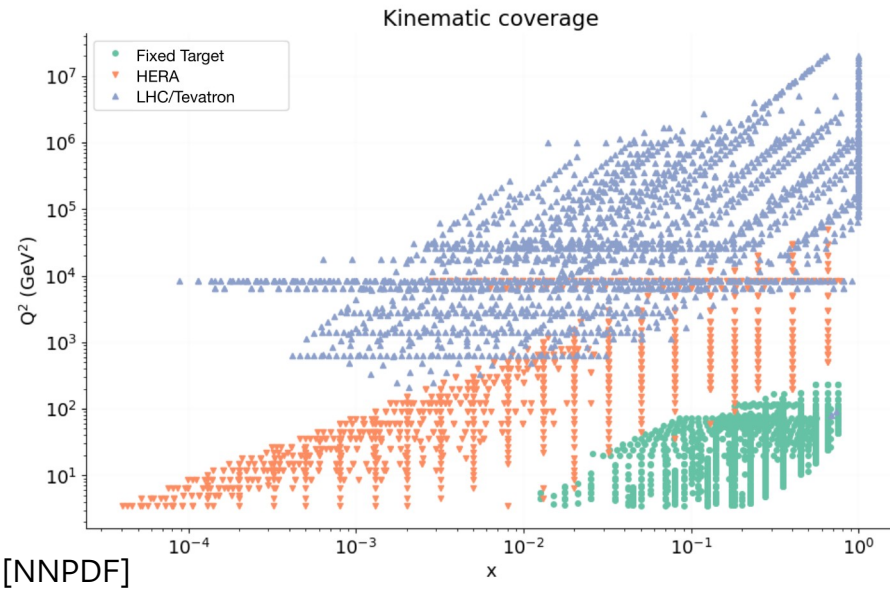
# How to determine PDFs

$$\sigma_{h_1 h_2 \to X} = \sum_{ij} \int_0^1 \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 f_{i,h_1}(x_1, \mu_F^2) f_{j,h_2}(x_2, \mu_F^2) \hat{\sigma}_{ij \to X}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2) + \mathcal{O}\left(\frac{\Lambda^2}{Q^2}\right)$$

Predicted in pQCD:

$$\hat{\sigma} = \hat{\sigma}^{(0)} + \hat{\sigma}^{(1)} +^2 \hat{\sigma}^{(2)} + \ldots$$

Measured




[NNPDF]

Ill-posed inversion problem! In practise:

1. define a parameterization:
   - polynomials
   - neutral networks
   - ...
2. extract parameters in a global fit

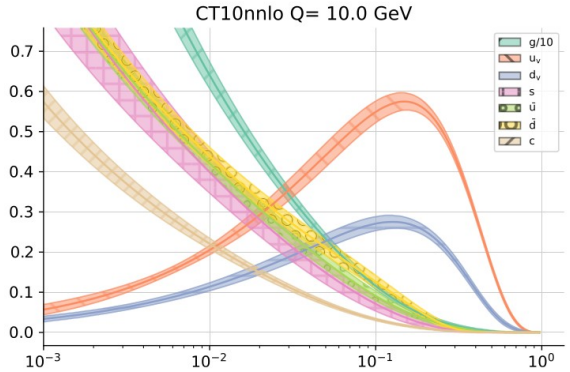
# Impact of global fits

DIS only fits
(from around 2010)

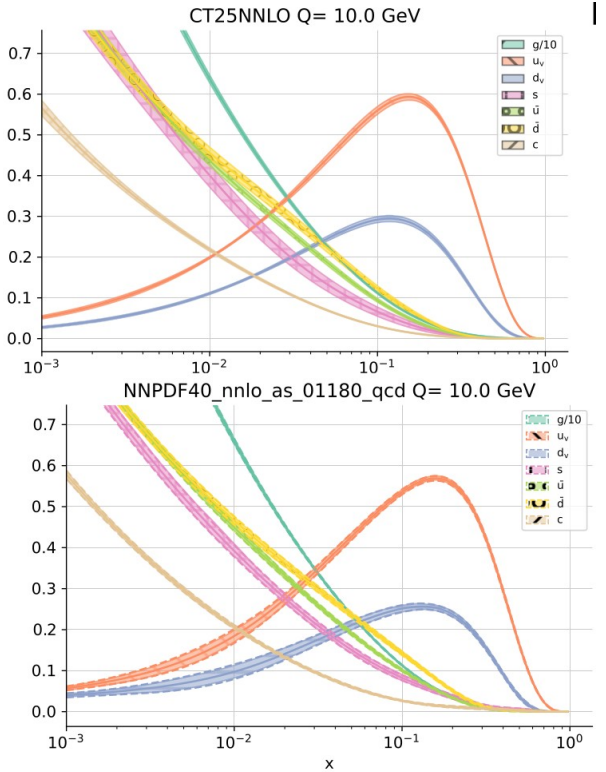
State-of-the-art

Gluon:

u-quark:




+ LHC data



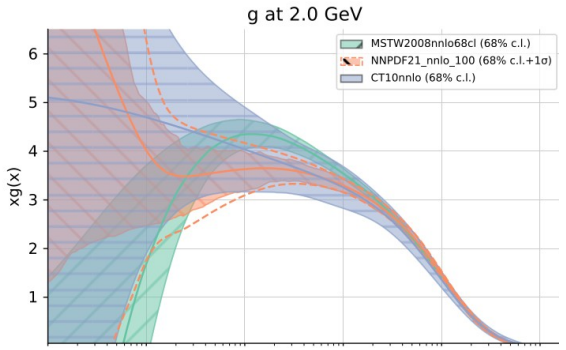

[Credit: Juan Cruz Martinez's DIS26 talk]

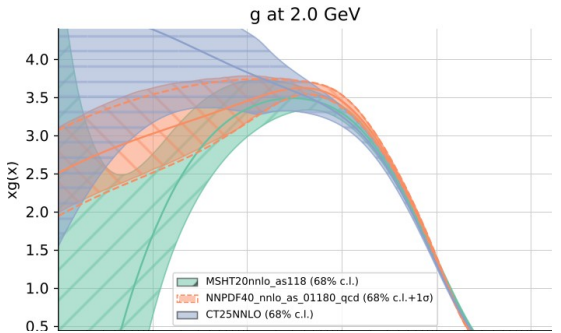
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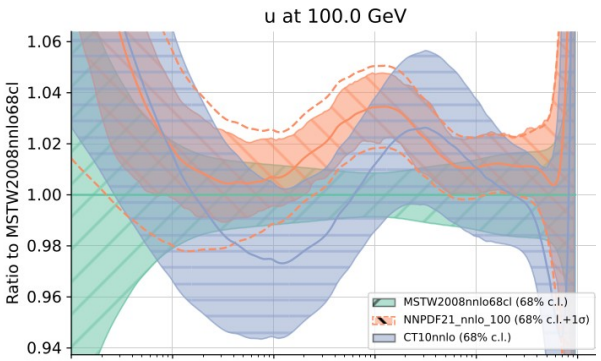
Gluon:

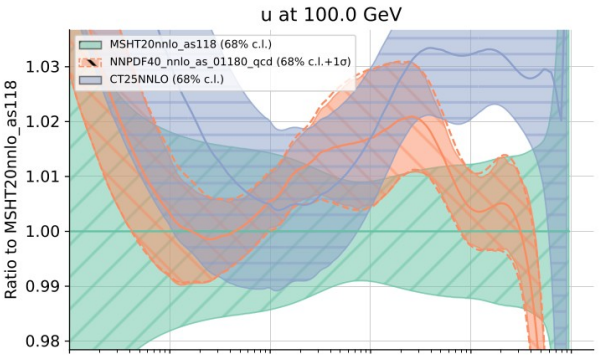



+ LHC data

u-quark:



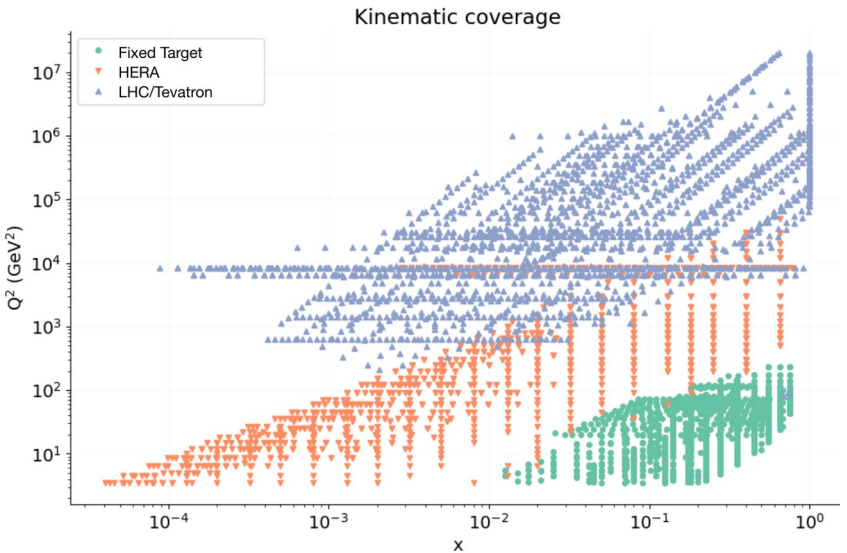

[Credit: Juan Cruz Martinez's DIS26 talk]

# How to determine PDFs: challenges

$$\sigma_{h_1 h_2 \to X} = \sum_{ij} \int_0^1 \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 f_{i,h_1}(x_1, \mu_F^2) f_{j,h_2}(x_2, \mu_F^2) \hat{\sigma}_{ij \to X}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2) + \mathcal{O}\left(\frac{\Lambda^2}{Q^2}\right)$$

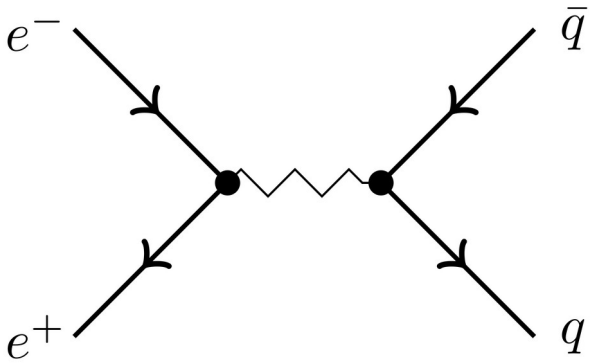
Many things go into these fits:

- Higher twist
- Resummation
- Heavy mass flavour scheme
- Parametrisation
- Fitting scale
- Treatment of missing higher order uncertainties
- Electroweak parameters + corrections
- Quark masses
- Evolution settings
- Fitting technique



# Anatomy of higher order QCD computations

# e+e- → jets: our higher-order QCD playground




$$a + b \to 1 + 2 + \cdots + n$$

Fermi's Golden Rule: $\mathrm{d}\sigma = \frac{1}{F}\langle|\mathcal{M}_n|^2\rangle \mathrm{d}\Phi_n$

Flux factor: $F = p_a \cdot p_b$

Lorentz Invariant Phase Space (LIPS):

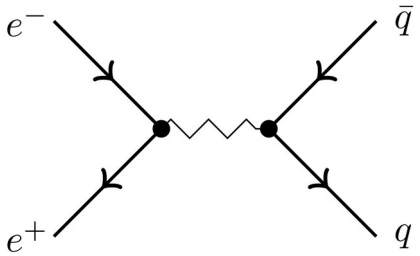
$$\mathrm{d}\Phi_n = \delta(p_a + p_b - \sum_{i=1}^{n} p_i) \prod_{i=1}^{n} \frac{\mathrm{d}^d p_i}{(2\pi)^d} 2\pi\theta(E_i)\delta(p_i^2 - m_i^2)$$

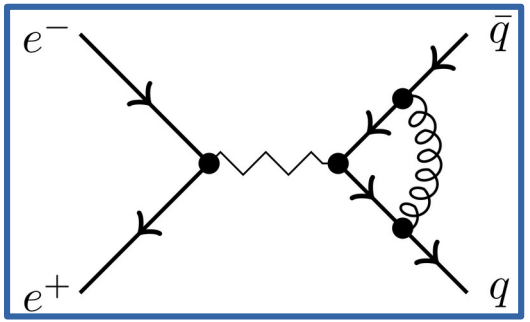
Leading order cross section:
(only photon exchange)

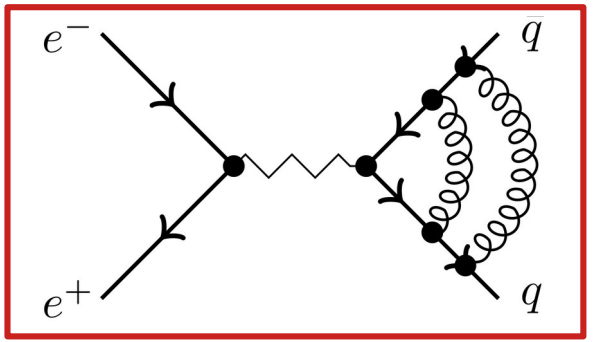
$$\sigma^{\mathrm{LO}}(e^+e^- \to q\bar{q}) = \frac{4\pi\alpha^2}{3s} N_c Q_q^2 \quad s = (p_a + p_b)^2$$

quark electric charge

# Perturbative expansion




$$|\mathcal{M}_2|^2 = |\mathcal{M}_2^{(0)}|^2 + 2\,\mathrm{Re}\left[\left(\mathcal{M}_2^{(0)}\right)^\dagger \mathcal{M}_2^{(1)}\right] + 2\,\mathrm{Re}\left[\left(\mathcal{M}_2^{(0)}\right)^\dagger \mathcal{M}_2^{(2)}\right] + |\mathcal{M}_2^{(1)}|^2 + \dots$$

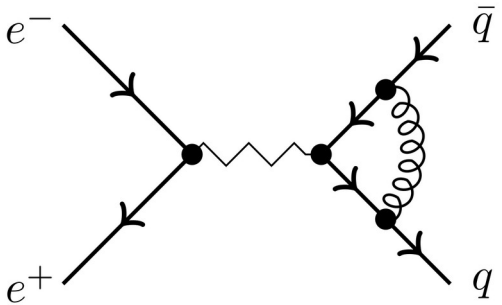
Next-to-leading order (NLO)

Next-to-next-to-leading order (NNLO)

$\mathcal{O}(\alpha^2)$ $\mathcal{O}(\alpha^2\alpha_s)$ $\mathcal{O}(\alpha^2\alpha_s^2)$

# e+e-: the virtual corrections

→ UV counter terms not needed at NLO QCD




$$\int \mathrm{d}^4 q \frac{\cdots}{q^2(q+p_1)^2(q-p_2)^2} \quad |q| \to 0 \sim \int_0 \frac{\mathrm{d}q}{q}$$

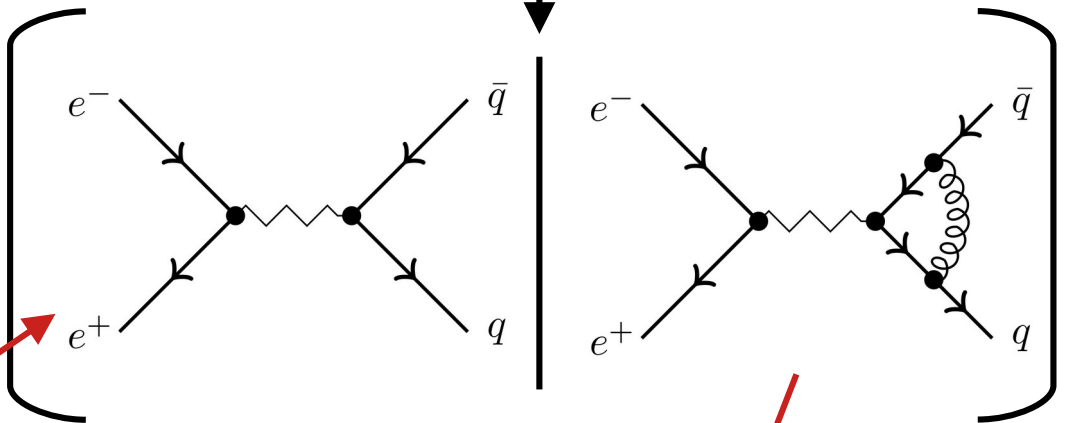
In dimensional regularization: $\sim \int_0 \frac{\mathrm{d}q}{q^{1+\varepsilon}}$

# e+e-: the virtual corrections

$$2\,\mathrm{Re}\left[\left(|\mathcal{M}_2^{(0)}\right)^\dagger |\mathcal{M}_2^{(1)}\right]$$

Sum over spin & color implied

$$2\,\mathrm{Re}$$




$$\mathcal{M}^\nu_{e^+e^-}\frac{-g_{\mu\nu}}{q^2}\mathcal{M}^\mu_{q\bar{q}} = \bar{v}(p_{e-})\,(-ie\gamma_\mu)\,u(p_{e^-})\frac{1}{q^2}\bar{u}(p_q)\,(-ie_q e\delta_{ab}\gamma^\mu)\,v(p_{\bar{q}})$$

$$\mathcal{M}^\mu_{q\bar{q}g} = \int \frac{\mathrm{d}^d p_g}{(2\pi)^d}\bar{u}(p_q)(-it^A_{ik}\gamma^\alpha)\frac{\not{p}_q - \not{p}_g}{(p_q-p_g)^2}\,(-ie_q e\delta_{km}\gamma^\mu)\,\frac{\not{p}_{\bar{q}} + \not{p}_g}{(p_{\bar{q}}+p_g)^2}(-it^B_{mj}\gamma^\beta)v(p_{\bar{q}})\frac{-i\delta^{AB}}{p_g^2}\left(g_{\beta\alpha} - \frac{p_g^\beta n^\alpha + p_g^\alpha n^\beta}{p_g n}\right)$$

# e+e-: the virtual corrections

Spin sum: $\sum_{\text{pol}} u_a(p)\bar{u}_b(p) = \not{p}_{ab}$

$$\operatorname{Tr}\left(\gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu\right) = d(g^{\kappa\lambda}g^{\mu\nu} - g^{\kappa\mu}g^{\lambda\nu} + g^{\kappa\nu}g^{\lambda\mu})$$
$$\operatorname{Tr}\gamma^\mu\gamma_\nu\gamma_\mu = (2-d)\gamma_\nu$$

Leptonic side: $\mathcal{M}^{\mu,\dagger}_{e^+e^-}\mathcal{M}^{\nu}_{e^+e^-} \sim \operatorname{Tr}\not{p}_{e^+}\gamma_\mu\not{p}_{e^-}\gamma_\nu$

Dirac part of $\mathcal{M}^{\mu,\dagger}_{q\bar{q}}\mathcal{M}^{\nu}_{q\bar{q}g}$ : $\operatorname{Tr}\left\{\not{p}_{\bar{q}}\gamma^\mu\not{p}_q\gamma^\alpha)(\not{p}_q - \not{p}_g)\gamma^\nu(\not{p}_{\bar{q}} + \not{p}_g)\gamma^\beta)\right\}$

$\ni p_q^\alpha p_{\bar{q}}^\beta \operatorname{Tr}\gamma^\mu\not{p}_q\gamma^\nu\not{p}_{\bar{q}}$ $\ni \operatorname{Tr}\left\{\not{p}_{\bar{q}}\gamma^\mu\not{p}_q\gamma^\alpha\not{p}_g\gamma^\nu\not{p}_g\gamma^\beta\right\}$

Terms with a single loop momentum vanish due to symmetry: $\int \mathrm{d}^d p_g p_g^\mu f(p_g^2) \to 0$

Colour sum: $\operatorname{Tr} t^A t^A = T_F\delta^{AA} = N_C C_F$

# Feynman integrals

After some work, we will be left with an expression in terms of $Q^2$ and Feynman integrals:

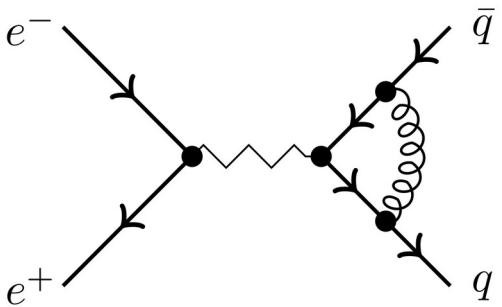
$$\int \frac{\mathrm{d}^d p_g}{(2\pi)^d} \frac{1}{(p_g + p_q)^2} \frac{1}{(p_g - p_{\bar{q}})^2} \frac{1}{p_g^2} \sim \left( \frac{\mu^2}{-q^2} \right)^{(d-4)/2} \qquad q^2 = (p_q + p_{\bar{q}})^2$$

Many techniques to evaluate Feynman integrals (3rd lecture):

- Feynman parameters
- Differential equations
- Numerical methods

# e+e-: the virtual corrections

→ UV counter terms not needed at NLO QCD

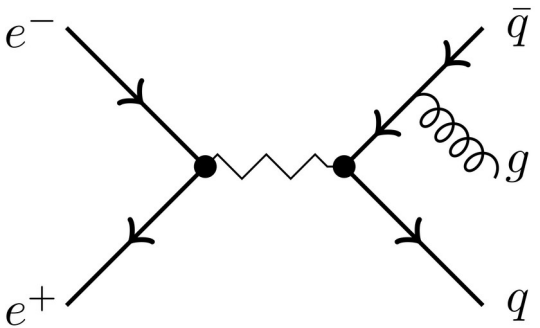



After some algebra:

$$\sigma^{\mathrm{LO}} \left[ \frac{\alpha_s}{2\pi} C_F \frac{1}{\Gamma(1-\varepsilon)} \mathrm{Re} \left( \frac{4\pi\mu^2}{-s-i0} \right)^{\varepsilon} \left( \frac{-2}{\varepsilon^2} - \frac{3}{\varepsilon} - 8 \right) \right]$$

# Real corrections

Consider the following matrix element:




$$|\mathcal{M}_3|^2 = |\mathcal{M}_3^{(0)}|^2 + 2\,\mathrm{Re}\left[\left(|\mathcal{M}_3^{(0)}\right)^\dagger |\mathcal{M}_3^{(1)}\right] + \dots$$

$\mathcal{O}(\alpha^2\alpha_s)$ $\mathcal{O}(\alpha^2\alpha_s^2)$

$$\rightarrow \quad \frac{4\pi\alpha^2}{3s} N_c Q_q^2 \int \mathrm{d}\Phi_3 g_s^2 C_F 2 \left(\frac{s_{13}+s_{23}+2s_{12}}{s s_{13} s_{23}}\right)$$

# Real corrections in 4 dimensions → IR limits exposed

A bit of phase space magic: $\delta^d(p_a + p_b - p_1 - p_2 - p_3)$

3n-4=5 phase space dimensions → Integrate out two angles

$$x_i = \frac{2p_i \cdot p}{s} = \frac{s_{ij} + s_{ik}}{s} \to s_{ij} = 1 - x_k \qquad x_1 + x_2 + x_3 = 2$$

$$\mathrm{d}\Phi_3 = \frac{s}{2(4\pi)^3}\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3\delta(2 - x_1 - x_2 - x_3)$$

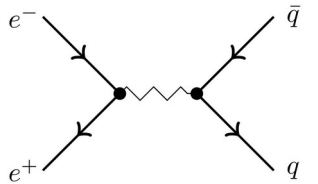
… reveals hidden treasure: $\sim \mathrm{d}\Phi_3 \left( \frac{x_q^2 + x_{\bar{q}}^2}{(1 - x_q)(1 - x_{\bar{q}})} \right)$

Singularities:
- collinear: $x_q \to 1$ or $x_{\bar{q}} \to 1$
- soft: $x_q \to 1$ and $x_{\bar{q}} \to 1$

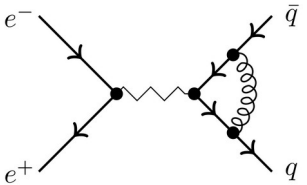
Regularize in dimensional regularization: $\sigma^{\mathrm{LO}} \left[ \frac{\alpha_s}{2\pi} C_F \frac{\Gamma(1-\varepsilon)^2}{\Gamma(1-3\varepsilon)} \left( \frac{4\pi\mu^2}{s} \right)^{\varepsilon} \left( \frac{2}{\varepsilon^2} + \frac{3}{\varepsilon} + \frac{19}{2} \right) \right]$

# Combined NLO QCD

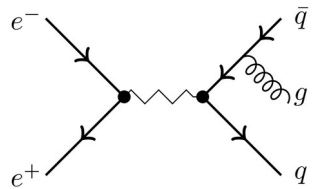
$$\sigma^{\mathrm{LO}}$$




- $$\sigma^{\mathrm{LO}}\left[\frac{\alpha_s}{2\pi}C_F\frac{1}{\Gamma(1-\varepsilon)}\,\mathrm{Re}\left(\frac{4\pi\mu^2}{-s-i0}\right)^{\varepsilon}\left(\frac{-2}{\varepsilon^2}-\frac{3}{\varepsilon}-8\right)\right]$$



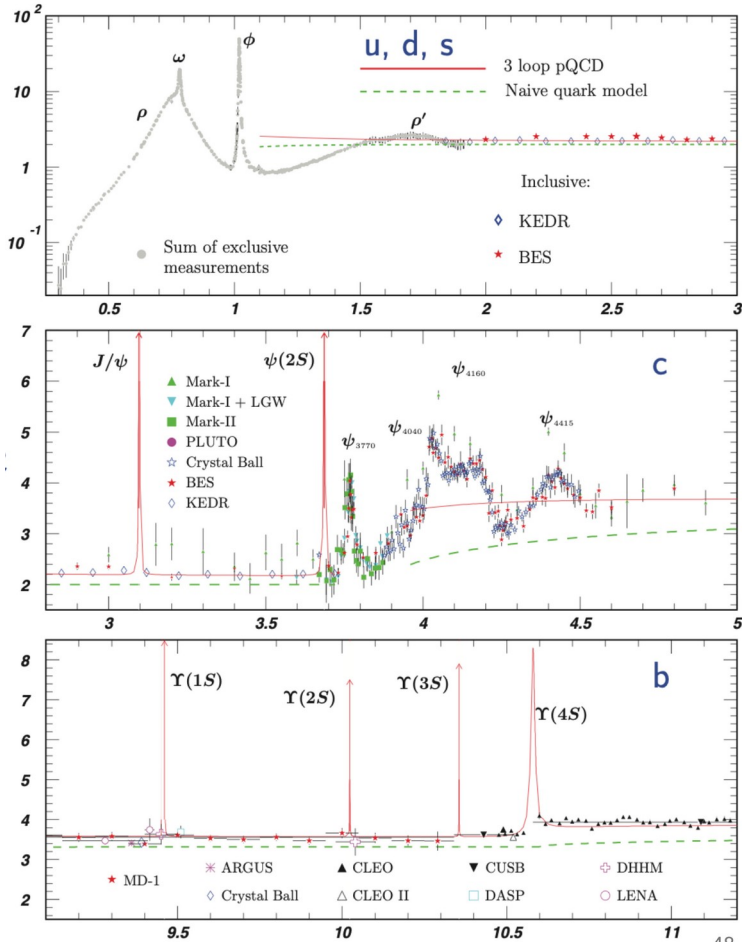

- $$\sigma^{\mathrm{LO}}\left[\frac{\alpha_s}{2\pi}C_F\frac{\Gamma(1-\varepsilon)^2}{\Gamma(1-3\varepsilon)}(\frac{4\pi\mu^2}{s})^{\varepsilon}\left(\frac{2}{\varepsilon^2}+\frac{3}{\varepsilon}+\frac{19}{2}\right)\right]$$




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$$\sigma^{\mathrm{NLO}} = \sigma^{\mathrm{LO}}(1+\frac{\alpha_s}{\pi})$$

# R-ratio




$$R = \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} = N_c \sum_q Q_q^2 \theta(\sqrt{s} - 2m_q)$$

At the Z-pole:
(this includes now also Z-boson exchange)

$$R^{\text{LO}} = 20.09$$

$$R^{\text{NLO}} = 20.89$$

$$R^{\text{LEP}} = 20.79 \pm 0.09$$

# Kinoshita-Lee-Nauenberg Theorem

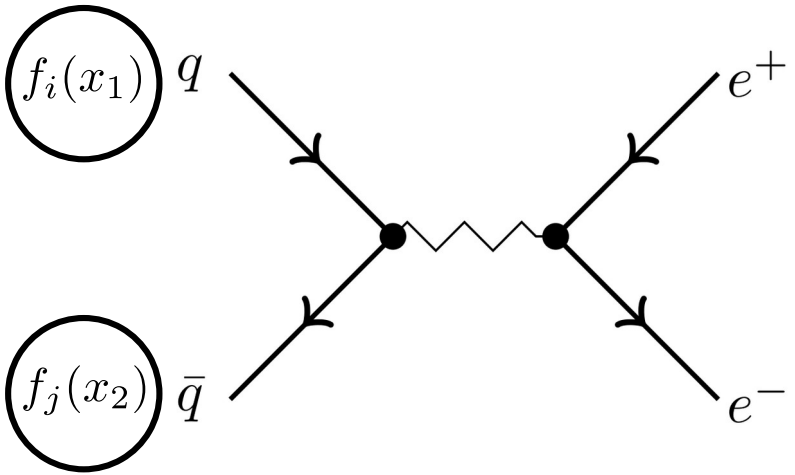
When calculating **sufficiently inclusive** observable quantities in quantum field theory (like cross sections or decay rates), **all infrared (soft and collinear) divergences cancel out** once you include every process that is **physically indistinguishable** within the detector resolution (i.e., summing over degenerate initial and final states).

This is ‘trivial’ for inclusive quantities like total cross sections,
but as soon we have to be careful as soon we want something more differential

Differential: not fully integrating over the phase space but only in some region of it

# NLO@hadron colliders: The Drell-Yan process

Simplest and probably best understood hadron-hadron process




Kinematics of the boson:

$$q^\mu = (p_1 + p_2)^\mu$$

$$Q = \sqrt{q^2}\,, \quad Y = \frac{1}{2}\ln\frac{q^0 + q^3}{q^0 - q^3}$$

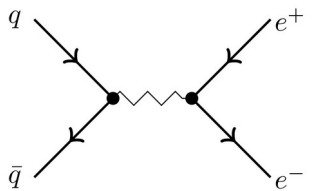
$$\xi_{i/j} = \frac{Q}{\sqrt{s}} e^{\pm Y}$$

$$\sigma_{pp\to e^+e^-} = \sum_{ij}\int_0^1\int_0^1 \mathrm{d}x_1\mathrm{d}x_2 f_i(x_1) f_j(x_2)\hat{\sigma}_{ij\to e^+e^-}(\alpha_s(\mu_R^2), \mu^2) + \mathcal{O}\left(\frac{\Lambda^2}{Q^2}\right)$$

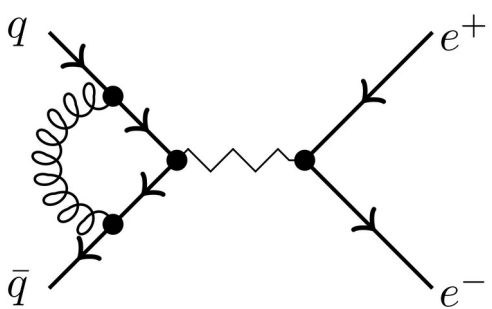
$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}Q\mathrm{d}Y} = \sum_{ij}\frac{1}{n_i n_j}\frac{Q}{\xi_i\xi_j s}\int_{\xi_i}^1\frac{\mathrm{d}z_i}{z_i} f_i(\frac{\xi_i}{z_i}, \mu_F^2)\int_{\xi_j}^1\frac{\mathrm{d}z_j}{z_j} f_j(\frac{\xi_j}{z_j}, \mu_F^2) F_{ij}(Q^2, z_i, z_j, \mu_R^2, \mu_F^2)$$

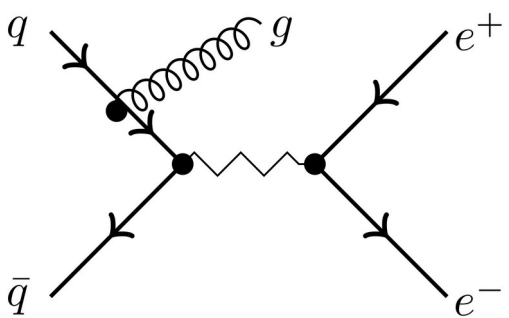
# Drell-Yan: Partonic channels

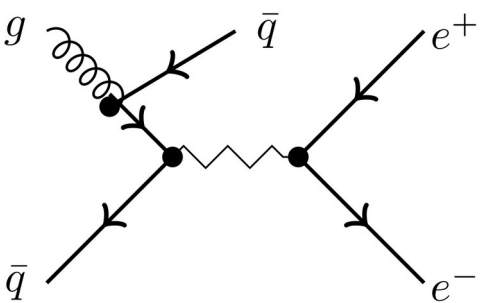
At leading order we have: $(ij) = (q\bar{q}), (\bar{q}q) \quad \text{for } q \in \{d, u, s, \dots\}$




At next-to-leading order we have: $(ij) = (q\bar{q}), (\bar{q}q), (qg), (gq), (\bar{q}g), (g\bar{q}) \quad \text{for } q \in \{d, u, s, \dots\}$



# Collinear initial state singularities

$q\bar{q}$ channel: LO: $F_{q\bar{q}}^{(0)} = \delta(1-z_i)\delta(1-z_j)\bar{F}_{q\bar{q}}^{0}(Q^2)$

$$F_{q\bar{q}}^{(1),\text{real}} = \frac{\alpha_s}{2\pi} C_F c_\varepsilon \left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} \left[\delta(1-z_i)\delta(1-z_j)\left(\frac{2}{\varepsilon^2}\right) - \delta(1-z_i)\frac{1}{\varepsilon}\left(\frac{1+z_j^2}{1-z_j}\right)_+ - (z_i \leftrightarrow z_j) + \text{finite}\right] \bar{F}_{q\bar{q}}^{(0)}$$

$$F_{q\bar{q}}^{(1),\text{virtual}} = \frac{\alpha_s}{2\pi} C_F c_\varepsilon \left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} \left[\delta(1-z_i)\delta(1-z_j)\left(-\frac{2}{\varepsilon^2} - \frac{3}{\varepsilon}\right) + \text{finite}\right]$$

$$F_{q\bar{q}}^{(1)} = \frac{\alpha_s}{2\pi} c_\varepsilon \left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} \frac{1}{\varepsilon} \left[\delta(1-z_i) P_{qq}(z_j) + (z_i \leftrightarrow z_j)\right] \bar{F}_{q\bar{q}}^{(0)}$$

**single poles do not cancel but ~ splitting functions….**

$qg$ channels:

$$F_{qg}^{(1),\text{real}} = \frac{\alpha_s}{2\pi} C_F c_\varepsilon \left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} \left[-\delta(1-z_i)\frac{1}{\varepsilon}\left(z_j^2 + (1-z_j)^2\right) + \text{finite}\right] \bar{F}_{q\bar{q}}^{(0)}$$

$$F_{qg}^{(1),\text{virtual}} = 0$$

$$F_{qg}^{(1)} = \frac{\alpha_s}{2\pi} \frac{N_c^2 - 1}{N_c} c_\varepsilon \left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} \left[-\delta(1-z_i)\frac{1}{\varepsilon} P_{qg}(z_j)\right] \bar{F}_{q\bar{q}}^{(0)}$$

$$c_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)}$$

# NLO PDFs

Absorb the singularity in the definition of the PDF → renormalised PDFs

$$\frac{1}{\varepsilon}\left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} = \frac{1}{\varepsilon} + \ln\left(\frac{\mu^2}{Q^2}\right) + \mathcal{O}(\varepsilon^2) = \underbrace{\frac{1}{\varepsilon} + \ln\left(\frac{Q^2}{\mu_F^2}\right)}_{\text{part of the PDF}} + \ln\left(\frac{\mu_F^2}{Q^2}\right) + \mathcal{O}(\varepsilon^2)$$

part of the partonic cross section

Renormalised PDFs

$$f_a(x) \to f_a(x,\mu_F^2) = f_a(x) + \sum_b \frac{\alpha_s}{2\pi} c_\varepsilon \left(\frac{\mu^2}{\mu_F^2}\right)^{\varepsilon} \frac{1}{\varepsilon} \int_x^1 P_{ab}(z) f_b(x/z)$$

Important: the PDF depends now also on the order of the computation
For example: NNPDF31_nnlo_as0118 vs. NNPDF31_nlo_as0118

# End of lecture 2 - summary

- Parton distribution functions
  - Evolution
  - Determination
- e+e- @ NLO QCD
  - Virtual corrections
  - Real corrections
  - Regulation and cancellation of IR divergencies
- DY & NLO PDFs